



Student Number: _____

GIRRAWEE HIGH SCHOOL

2024

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen.
- NESA approved calculators may be used.
- A reference sheet is provided.
- In section II, show relevant mathematical reasoning and/or calculations.

**Total marks:
70**

Section I – 10 marks

- Attempt questions 1 - 10
- Allow about 15 minutes for this section.

Section II – 60 marks

- Attempt questions 11 - 16.
- Allow about 1 hour and 45 minutes for this section.
- Start each question on a new page.

Section I

10 marks

Attempt questions 1 - 10

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for questions 1-10

1. Given that $\sin A = \frac{5}{13}$ and $\sin B = \frac{8}{17}$, find the value of $\sin(A + B)$. A and B are acute angles.

A. $-\frac{107}{140}$

B. $-\frac{21}{221}$

C. $\frac{107}{140}$

D. $\frac{171}{221}$

2. The angle between the vectors $\underline{u} = -2\underline{i} + 6\underline{j}$ and $\underline{v} = 4\underline{i} - 2\underline{j}$ is closest to:

A. 45°

B. 60°

C. 135°

D. 120°

3. What is the exact value of $\int_0^{\frac{\pi}{8}} \cos^2 x \, dx$?

A. $\frac{\pi - 2\sqrt{2}}{8}$

B. $\frac{\pi - 2\sqrt{2}}{16}$

C. $\frac{\pi + 2\sqrt{2}}{8}$

D. $\frac{\pi + 2\sqrt{2}}{16}$

4. What is the value of $\int_0^1 x^2 \sqrt{1-x^3} \, dx$, using the substitution $u = 1 - x^3$?
- A. $-\frac{2}{9}$
- B. $-\frac{1}{9}$
- C. $\frac{1}{9}$
- D. $\frac{2}{9}$
5. A course regulation requires that the same number of students achieve each grade from A to E where possible.
What is the smallest number of students required to ensure that at least one particular grade is awarded 7 times?
- A. 30
- B. 31
- C. 35
- D. 36
6. A circular disc is heated so that the face is increasing in area at a constant rate of $10 \text{ cm}^2/\text{s}$. What is the rate at which the radius of the disc is increasing when the radius is 6 cm?
- A. 0.26 cm/s
- B. 0.27cm/s
- C. 3.76 cm/s
- D. 3.77 cm/s

7. What is the multiplicity of the root $x = 1$ of the equation

$$f(x) = 3x^5 - 5x^4 + 5x - 3 ?$$

- A. 1
- B. 2
- C. 3
- D. 4

8. What is the inverse function of $f(x) = e^{x^3}$?

- A. $f^{-1}(x) = 3e^x$
- B. $f^{-1}(x) = 3\ln x$
- C. $f^{-1}(x) = \sqrt[3]{\ln x}$
- D. $f^{-1}(x) = \sqrt{3\ln x}$

9. A curve is represented by the following parametric equations:

$$x = 2 \cos \theta, y = 2 \sin \theta$$

Which of the following is the cartesian equation of the curve?

- A. $x + y = 2$
- B. $x + y = 4$
- C. $x^2 + y^2 = 2$
- D. $x^2 + y^2 = 4$

10. The polynomial $f(x) = 2x^2 + kx + 4$ can be expressed as $f(x) = (x - 2)g(x) + 6$. Which of the following is the correct expression for $g(x)$?
- A. $2x - 1$
 - B. $2x + 1$
 - C. $2x - 3$
 - D. $2x + 3$

Section II

60 marks

Attempt all questions

Allow about 1 hour and 45 minutes for this section.

Start each question on a new page in the answer booklet provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing paper is available on request.

Question 11 (10 marks)

- a. Solve $\frac{1}{2x-1} - x > 0$ [3]
- b. The polynomial $4x^3 - 12x^2 + 5x + 6 = 0$ has roots α , β and γ .
Find α , β and γ given that one of the roots is the sum of the other two. [3]
- c. (i) Write $3 \cos x - \sqrt{3} \sin x$ in the form $R \cos(x + \alpha)$. [2]
- (ii) Hence, solve $3 \cos x - \sqrt{3} \sin x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. [2]

Question 12 (10 marks)

- a. Given that $\underline{u} = 2\underline{i} + 3\underline{j}$ and $\underline{v} = -2\underline{i} + 4\underline{j}$, find $\text{proj}_{\underline{u}} \underline{v}$. [2]
- b. Use the substitution $t = \tan \frac{x}{2}$ to solve
 $\cos x - 3 \sin x + 3 = 0$ for $0 \leq x \leq 2\pi$. [4]
- c. Use the substitution $u = \sin x$ to find $\int -\cos x \sin^{\frac{3}{2}} x \, dx$. [2]
- d. In how many ways can 8 people be placed in groups of 2? [2]

Question 13 (10 marks)

- a. Factorise $P(x) = 4x^3 - 3x^2 - 25x - 6$ [3]
- b. Prove that $\sec A (1 - \sin A)(\sec A + \tan A) = 1$ [2]
- c. Use mathematical Induction to prove that $3^{3n} + 2^{n+2}$ is divisible by 5
for $n \geq 1$. [3]
- d. Find $\int \frac{1}{9 + 16x^2} \, dx$ [2]

Question 14 (10 marks)

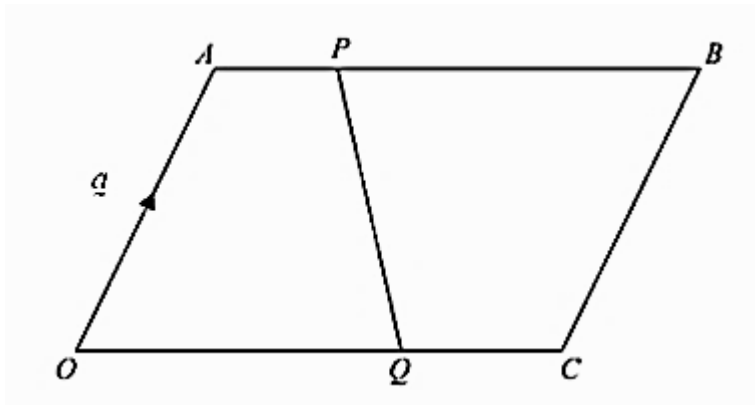
- a. In the diagram below, $OABC$ is a parallelogram. [3]

$$\overrightarrow{OA} = \underline{\underline{a}} \text{ and } \overrightarrow{OC} = \underline{\underline{c}}.$$

P is the point on AB such that $AP = \frac{1}{4} AB$.

Q is the point on OC such that $OQ = \frac{2}{3} OC$.

Find $\overrightarrow{PQ}$ in terms of $\underline{\underline{a}}$ and $\underline{\underline{c}}$, giving your answer in simplest surd form.



- b. Newton's law of cooling states that the rate of change of the temperature θ , of a body at any time t , is proportional to the difference in the temperature of the body and the temperature m , of the surrounding medium, i.e.

$$\frac{d\theta}{dt} = k(\theta - m), \text{ where } k \text{ is a constant.}$$

- (i) Show that $\theta = m + Ae^{kt}$ where A is a constant, satisfies this equation. [1]

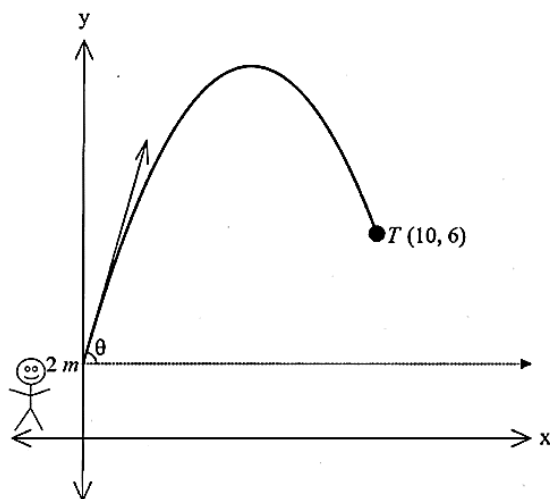
- (ii) If the temperature of the surrounding air is $40^\circ C$ and the temperature of the body drops from $170^\circ C$ to $105^\circ C$ in 45 minutes, find the temperature of the body in another 90 minutes (to 2 decimal places). [2]

- (iii) Find the time taken for the temperature of the body to drop to $80^\circ C$ (to the nearest minute). [2]

- c. Find the coefficient of x in the expansion of $\left(x + \frac{2}{x^2}\right)^{10}$ [2]

Question 15 (10 marks)

- a. David throws a pebble from a height of 2 metres at an angle of θ to the horizontal, with a velocity of 30m/s.



- (i) Show that the expression for the horizontal and vertical displacements at t seconds after projection are $x = 30t \cos \theta$ and $y = -5t^2 + 30t \sin \theta + 2$ respectively. [2]
(Take $g = 10\text{m/s}$ and take the origin to be the ground directly below David)
 - (ii) Show that the equation of the path of the particle is:
$$y = \frac{-x^2}{180} (1 + \tan^2 \theta) + x \tan \theta + 2$$
 [2]
 - (iii) If David manages to hit a target at point T , which is 10 metres away horizontally and 6 metres high, find the two possible angles of projection. [2]
- b. For the function $y = 2 \sin^{-1} (2x - 1) + \frac{\pi}{2}$
- (i) Find the domain and range. [2]
 - (ii) Sketch the curve, clearly showing the coordinates of the endpoints. [2]

Question 16 (10 marks)

- a. Solve the equation $\sin 3x - \sin x + \cos 2x = 0$ for $0 \leq x \leq 2\pi$. [3]
- b. (i) Prove that $8 \cos^4 x = 3 + 4 \cos 2x + \cos 4x$. [2]
- (ii) Hence, find the volume generated when the area bounded by the curve $y = \cos^2 x$, the x -axis and $x = 0$ and $x = \frac{\pi}{4}$ is rotated about the x -axis. [2]
- c. A committee of 5 is to be chosen from six men and seven women.
- (i) How many committees are possible if there are no restrictions? [1]
- (ii) How many committees are possible if there are more women than men? [2]

End of Examination

Mathematics Assessment Solutions
Girraween High School

Task: Trial HSC Extension 1

Year: 2024

Suggested Solutions

Question 1-2

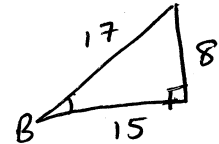
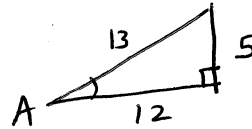
1. D 2. C 3. D 4. D 5. B 6. B 7. C 8. C 9. D 10. B

1. A and B lie in Q1

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$= \frac{5}{13} \times \frac{15}{17} + \frac{12}{13} \times \frac{8}{17}$$

$$= \frac{171}{221} \quad [D]$$



$$2. \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$= \frac{(-2) \times 4 + 6 \times (-2)}{\sqrt{40} \times \sqrt{20}}$$

$$= \frac{-20}{20\sqrt{2}}$$

$$\theta = 135^\circ \quad [C]$$

Marker's Feedback

Mathematics Assessment Solutions
Girraween High School

Task: Trial HSC Extension 1

Year: 2024

Suggested Solutions

Question 3

$$\begin{aligned} 3. \quad & \int_0^{\pi/8} \cos^2 x \, dx \\ &= \int_0^{\pi/8} \frac{1}{2} (1 + \cos 2x) \, dx \\ &= \frac{1}{2} \left[x + \frac{1}{2} \sin 2x \right]_0^{\pi/8} \\ &= \frac{1}{2} \left[\frac{\pi}{8} + \frac{1}{2} \sin\left(\frac{\pi}{4}\right) - 0 \right] \\ &= \frac{\pi}{16} + \frac{\sqrt{2}}{8} \\ &= \frac{\pi + 2\sqrt{2}}{16} \quad [D] \end{aligned}$$

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Mathematics Assessment Solutions
Girraween High School

Task:

Year: 2024

Suggested Solutions

Question 4

$$\int_0^1 x^2 \sqrt{1-x^3} \, dx$$

$$= \frac{1}{3} \int_1^0 \sqrt{u} \, du$$

$$= \frac{1}{3} \left[\frac{2}{3} u^{3/2} \right]_0^1$$

$$= \frac{2}{9} \quad [D]$$

$$u = 1 - x^3$$

$$\frac{du}{dx} = -3x^2$$

$$dx = \frac{-1}{3x^2} du$$

$$\text{when } x=0, u=1$$

$$\text{when } x=1, u=0$$

Question 5

$$5 \times 6 + 1 = 31 \quad [B]$$

Marker's Feedback

Suggested Solutions

Question 6

$$\frac{dA}{dt} = 10 \text{ cm}^2/\text{s} ; \frac{dr}{dt} = ? ; r=6 ; A = \pi r^2$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$10 = 2\pi r \times \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{10}{2\pi r}$$

$$= \frac{10}{2\pi \times 6} \quad (\text{when } r=6)$$

$$\div 0.27 \text{ cm/s}$$

[B]

Question 7

$$P(x) = 3x^5 - 5x^4 + 5x - 3 ; P(1) = 0$$

$$P'(x) = 15x^4 - 20x^3 + 5 ; P'(1) = 0$$

$$P''(x) = 60x^3 - 60x^2 ; P''(1) = 0$$

$$P'''(x) = 180x^2 - 120x ; P'''(1) = 60 \neq 0$$

$$\therefore \text{Multiplicity} = 3$$

[C]

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Suggested Solutions

Question 8

$$f(x): y = e^{x^3}$$

$$f^{-1}(x): x = e^{y^3}$$

$$\ln x = \ln e^{y^3}$$

$$\ln x = y^3$$

$$y = \sqrt[3]{\ln x}$$

[C]

Question 9

$$\begin{aligned} x^2 + y^2 &= (2\cos\theta)^2 + (2\sin\theta)^2 \\ &= 4(\sin^2\theta + \cos^2\theta) \end{aligned}$$

$$x^2 + y^2 = 4$$

[D]

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Suggested Solutions

Question 10

$$f(x) = (x-2)g(x) + 6 \Rightarrow f(2) = 6 \quad (\text{Remainder theorem})$$

$$f(2) = 2 \times 2^2 + 2k + 4 = 6$$

$$12 + 2k = 6$$

$$2k = -6$$

$$k = -3$$

$$2x^2 - 3x + 4 = (x-2)g(x) + 6$$

$$2x^2 - 3x - 2 = (x-2)g(x)$$

$$(x-2)(2x+1) = (x-2)g(x)$$

$$\therefore g(x) = 2x+1$$

[B]

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Suggested Solutions

Question 11

a) $\frac{1}{2x-1} - x > 0, x \neq \frac{1}{2}$

Solve

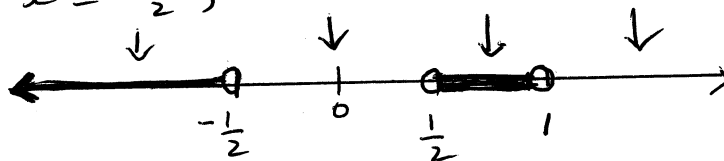
$$\frac{1}{2x-1} = x$$

$$2x^2 - x = 1$$

$$2x^2 - x - 1 = 0$$

$$(2x+1)(x-1) = 0$$

$$x = -\frac{1}{2}, 1$$



Test $x = -1$

$$\frac{1}{2(-1)-1} - (-1) > 0$$

✓

Test $x = 0$

$$\frac{1}{-1} - 0 > 0$$

✗

Test $x = 0.8$

$$\frac{1}{2(0.8)-1} - 0.8 > 0$$

✓

Test $x = 2$

$$\frac{1}{2(2)-1} - 2 > 0$$

✗

Solution :

$$x < -\frac{1}{2}, \frac{1}{2} < x < 1$$

3

Marker's Feedback

Suggested Solutions

Question 11

a)

$$\frac{1}{2x-1} - x > 0 \quad ; \quad x \neq \frac{1}{2}$$

$$\frac{1}{2x-1} > x$$

$$(x(2x-1)^2)$$

$$2x-1 > x(2x-1)^2$$

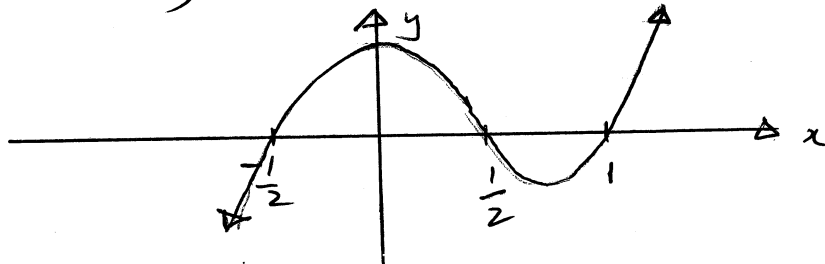
$$x(2x-1)^2 < 2x-1$$

$$x(2x-1)^2 - (2x-1) < 0$$

$$(2x-1) [x(2x-1) - 1] < 0$$

$$(2x-1) (2x^2 - x - 1) < 0$$

$$(2x-1) (2x+1) (x-1) < 0$$



$$\text{Solution : } x < -\frac{1}{2} , \quad \frac{1}{2} < x < 1$$

3

Marker's Feedback

Suggested Solutions

Question 1)

$$b) \quad 4x^3 - 12x^2 + 5x + 6, \text{ roots } \alpha, \beta, \gamma$$

$$\Rightarrow \alpha, \beta, \alpha + \beta$$

$$\text{Sum of roots: } \alpha + \beta + \alpha + \beta = 3$$

$$\alpha + \beta = \frac{3}{2}$$

$$\text{i.e. } \gamma = \frac{3}{2} \Rightarrow \alpha + \beta = \frac{3}{2} \quad \underline{1}$$

$$\text{Product of roots: } \frac{3}{2} \alpha \beta = -\frac{3}{2}$$

$$\alpha \beta = -1 \quad ; \quad \beta = \frac{3}{2} - \alpha$$

$$\text{Substitute } \beta = \frac{3}{2} - \alpha \text{ into } \alpha \beta = -1$$

$$\alpha \left(\frac{3}{2} - \alpha \right) = -1 \quad \underline{1}$$

$$\frac{3}{2} \alpha - \alpha^2 = -1$$

$$2\alpha^2 - 3\alpha - 2 = 0$$

$$(\alpha - 2)(2\alpha + 1) = 0$$

$$\alpha = 2, -\frac{1}{2}$$

$$\text{Roots: } \alpha = 2, \beta = -\frac{1}{2}, \gamma = \frac{3}{2} \quad \underline{1}$$

3

Marker's Feedback

Suggested Solutions

Question 11 c)

$$i) 3 \cos x - \sqrt{3} \sin x \equiv R \cos(x + \alpha)$$

$$= R \cos x \cos \alpha - R \sin x \sin \alpha$$

Equating like terms,

$$R \cos \alpha = 3 \text{ --- ①} ; R \sin \alpha = \sqrt{3} \text{ --- ②}$$

Squaring and adding ① and ②

$$R^2 (\sin^2 \alpha + \cos^2 \alpha) = 12$$

$$R = 2\sqrt{3} \quad (R > 0)$$

$$\sin \alpha = \frac{1}{2} ; \cos \alpha = \frac{\sqrt{3}}{2} \Rightarrow \alpha \text{ in } Q1$$

$$\therefore \alpha = \frac{\pi}{6}$$

$$\therefore 3 \cos x - \sqrt{3} \sin x \equiv 2\sqrt{3} \cos(x + \frac{\pi}{6}) \quad \boxed{2}$$

$$ii) 3 \cos x - \sqrt{3} \sin x = \sqrt{3}$$

$$2\sqrt{3} \cos(x + \frac{\pi}{6}) = \sqrt{3}$$

$$\cos(x + \frac{\pi}{6}) = \frac{1}{2}$$

$$x + \frac{\pi}{6} = \frac{\pi}{3} \text{ or } \frac{5\pi}{3}$$

$$x = \frac{\pi}{6}, \frac{3\pi}{2}$$

Marker's Feedback

Suggested Solutions

Question 12

a) $\underline{u} = 2\underline{i} + 3\underline{j}$; $\underline{v} = -2\underline{i} + 4\underline{j}$

$$\text{proj}_{\underline{u}} \underline{v} = \frac{\underline{u} \cdot \underline{v}}{\underline{u} \cdot \underline{u}} \cdot \underline{u} \quad //$$

$$= \frac{2 \times (-2) + 3 \times 4}{2^2 + 3^2} \cdot 2\underline{i} + 3\underline{j}$$

$$= \frac{8}{13} (2\underline{i} + 3\underline{j}) \quad //$$

2

Marker's Feedback

Suggested Solutions

Question 12

$$b) \quad t = \tan \frac{x}{2}$$

$$\cos x - 3 \sin x + 3 = 0$$

$$\frac{1-t^2}{1+t^2} - 3 \left(\frac{2t}{1+t^2} \right) + 3 = 0$$

$$1-t^2 - 6t + 3(1+t^2) = 0$$

$$1-t^2 - 6t + 3 + 3t^2 = 0$$

$$2t^2 - 6t + 4 = 0$$

$$t^2 - 3t + 2 = 0$$

$$(t-1)(t-2) = 0$$

$$t = 1, 2$$

$$\text{ie. } \tan \frac{x}{2} = 1 \quad \text{or} \quad \tan \frac{x}{2} = 2$$

$$\frac{x}{2} = \frac{\pi}{4}$$

$$\frac{x}{2} = 1.107$$

$$x = 2.21 \text{ (to 2dp)}$$

$$x = \frac{\pi}{2}$$

Test for $x = \pi$

$$\cos \pi - 3 \sin \pi + 3$$

$$= -1 + 3 \neq 0$$

$\therefore x = \pi$ is not a solution

$$\therefore x = \frac{\pi}{2}, 2.21$$

4

Marker's Feedback

Suggested Solutions

Question 12

$$c) \int -\cos x \sin^{3/2} x \, dx \quad ; \quad u = \sin x$$

$$du = \cos x \, dx$$

$$= -\int u^{3/2} \, du \quad //$$

$$= -\frac{u^{5/2}}{5/2} + C$$

$$= -\frac{2}{5} \sin^{5/2} x + C \quad //$$

2

$$d) \frac{{}^8C_2 x^6 {}^6C_2 x^4 {}^4C_2 x^2 {}^2C_2}{4!} \quad //$$

$$= 105 \quad //$$

Marker's Feedback

Suggested Solutions

Question 13

$$a) P(x) = 4x^3 - 3x^2 - 25x - 6$$

Try: Factors of $-6 = \pm 1, \pm 2, \pm 3, \pm 6$

$$P(-2) = 4(-2)^3 - 3(-2)^2 - 25(-2) - 6$$

$$= 0$$

$\therefore (x+2)$ is a factor

$$\begin{array}{r} 4x^2 - 11x - 3 \\ x+2 \overline{) 4x^3 - 3x^2 - 25x - 6} \\ \underline{- 4x^3 + 8x^2} \\ -11x^2 - 25x \\ \underline{- 11x^2 - 22x} \\ -3x - 6 \\ \underline{- 3x - 6} \\ 0 \end{array}$$

$$\therefore P(x) = (x+2)(4x^2 - 11x - 3)$$

$$= (x+2)(x-3)(4x+1)$$

3

Marker's Feedback

Suggested Solutions

Question 13

$$b) \sec A (1 - \sin A) (\sec A + \tan A) = 1$$

$$\text{LHS} = \sec A (1 - \sin A) (\sec A + \tan A)$$

$$= \frac{1}{\cos A} (1 - \sin A) \left(\frac{1}{\cos A} + \frac{\sin A}{\cos A} \right)$$

$$= \frac{1}{\cos A} (1 - \sin A) \frac{1}{\cos A} (1 + \sin A) \quad \underline{\underline{1}}$$

$$= \frac{1}{\cos^2 A} (1 - \sin^2 A)$$

$$= \frac{1}{\cos^2 A} (\cos^2 A) \quad \underline{\underline{1}}$$

$$= 1$$

$$= \text{RHS}$$

2

Marker's Feedback

Suggested Solutions

Question 13

c) $3^{3n} + 2^{n+2}$ is divisible by 5 ; $n \geq 1$

Step 1 : Show true for $n=1$

$$3^3 + 2^3 = 35 \text{ which is divisible by 5.}$$

Step 2 : Assume true for $n=k$ for integer $k > 0$

$$\text{i.e. } 3^{3k} + 2^{k+2} = 5M ; M \text{ is a positive integer}$$

Step 3 : Prove true for $n=k+1$

$$\text{i.e. } 3^{3k+3} + 2^{k+3} = 5N, N \text{ is a positive integer}$$

$$\text{LHS} = 3^{3k+3} + 2^{k+3}$$

$$= 3^{3k} \times 3^3 + 2^{k+2} \times 2$$

$$= (5M - 2^{k+2}) \times 27 + 2^{k+2} \times 2 \quad (\text{from assumption})$$

$$= 135M - 27 \times 2^{k+2} + 2 \times 2^{k+2}$$

$$= 135M - 25 \times 2^{k+2}$$

$$= 5(27M - 5 \times 2^{k+2})$$

$$= 5N \quad \therefore \text{If true for } n=k, \text{ then also true for } n=k+1$$

Step 4 : By the principle of Mathematical Induction, the result is true for all $n \geq 1$ [3]

Marker's Feedback

Suggested Solutions

Question 13

$$d) \int \frac{1}{9 + 16x^2} dx$$

$$= \int \frac{1}{3^2 + (4x)^2} dx$$

$$= \frac{1}{4} \int \frac{4}{3^2 + (4x)^2} dx \quad \underline{\underline{!}}$$

$$= \frac{1}{4} \times \frac{1}{3} \tan^{-1} \left(\frac{4x}{3} \right) + C$$

$$= \frac{1}{12} \tan^{-1} \left(\frac{4x}{3} \right) + C \quad \underline{\underline{!}}$$

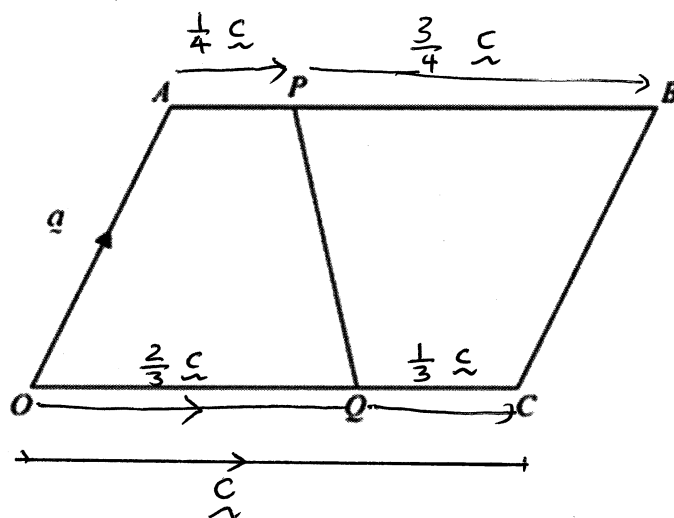
2

Marker's Feedback

Suggested Solutions

Question 14

a)



$$\vec{PQ} = \vec{PA} + \vec{AB} + \vec{OQ}$$

$$= -\frac{1}{4}\underline{c} - \underline{a} + \frac{2}{3}\underline{c}$$

$$= \left(\frac{2}{3} - \frac{1}{4}\right)\underline{c} - \underline{a}$$

$$= \frac{5}{12}\underline{c} - \underline{a}$$

||

||

||

3

Marker's Feedback

Suggested Solutions

Question 14

$$b) i) \theta = m + Ae^{kt} \quad \Rightarrow \quad Ae^{kt} = \theta - m$$

$$\frac{d\theta}{dt} = kAe^{kt}$$

$$= k(\theta - m) \quad (\text{since } Ae^{kt} = \theta - m)$$

!

$\therefore \theta = m + Ae^{kt}$ satisfies the equation.

$$ii) \theta = m + Ae^{kt}$$

At $t=0$,

$$170 = 40 + Ae^0$$

$$\therefore A = 130$$

$$\theta = 40 + 130e^{kt}$$

At $t=45$

$$105 = 40 + 130e^{45k}$$

$$65 = 130e^{45k}$$

$$45k \ln e = \ln \frac{65}{130}$$

$$45k = \ln \frac{65}{130} \quad !$$

$$k \doteq -0.0154$$

cont ... $\rightarrow$ P19

Marker's Feedback

Suggested Solutions

Question 14

b (ii) cont :

$$\text{At } t = 135$$

$$\theta = 40 + 130e^{135k}$$

$$= 56.25^{\circ}\text{C} \quad (\text{to 2dp})$$

!

$$\text{iii) } \theta = 80^{\circ}\text{C}, \quad t = ?$$

$$80 = 40 + 130e^{kt}$$

$$40 = 130e^{kt}$$

!

$$t = \ln \frac{4}{13} \div k$$

$$= 76.53$$

!

$$= 77 \text{ minutes}$$

2

Marker's Feedback

Suggested Solutions

Question 14

$$c) \left(x + \frac{2}{x^2} \right)^{10}$$

$$T_{k+1} = {}^nC_k a^{n-k} b^k$$

$$= {}^{10}C_k \cdot x^{10-k} \cdot (2x^{-2})^k$$

$$= {}^{10}C_k \cdot x^{10-k} \cdot 2^k x^{-2k}$$

$$= {}^{10}C_k \cdot 2^k \cdot x^{10-3k}$$

$$\text{Coefficient of } x \Rightarrow 10-3k=1$$
$$k=3$$

$$\therefore \text{Coefficient of } x = {}^{10}C_3 \cdot 2^3$$

$$= 960$$

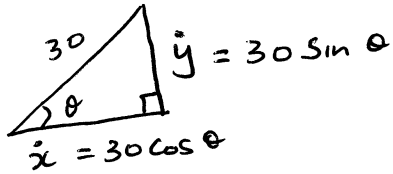
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Suggested Solutions

Question 15

a) i) Initially,

Resolving Forces,
Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = c_1$$

$$\text{When } t=0, \dot{x} = 30 \cos \theta$$

$$c_1 = 30 \cos \theta$$

$$\therefore \dot{x} = 30 \cos \theta$$

$$x = 30t \cos \theta + c_2$$

$$\text{When } t=0, x=0$$

$$\therefore c_2 = 0$$

$$x = 30t \cos \theta \quad \text{--- (1)}$$

Vertically

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c_1$$

$$\text{When } t=0, \dot{y} = 30 \sin \theta$$

$$c_1 = 30 \sin \theta$$

$$\therefore \dot{y} = -10t + 30 \sin \theta$$

$$y = -5t^2 + 30t \sin \theta + c_2$$

$$\text{When } t=0, y=2$$

$$\therefore c_2 = 2$$

$$y = -5t^2 + 30t \sin \theta + 2 \quad \text{--- (2)}$$

2

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Suggested Solutions

Question 15

a) (ii) From ①

$$t = \frac{x}{30 \cos \theta} \quad \text{--- ③}$$

substituting ③ into ②

$$y = -5 \left(\frac{x}{30 \cos \theta} \right)^2 + 30 \sin \theta \left(\frac{x}{30 \cos \theta} \right) + 2 \quad \quad \quad \parallel$$

$$= \frac{-5}{30^2} x^2 \sec^2 \theta + x \tan \theta + 2$$

$$= \frac{-x^2}{180} (1 + \tan^2 \theta) + x \tan \theta + 2 \quad \quad \quad \parallel$$

2

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Suggested Solutions

Question 15

a) (iii)

when $x=10$, $y=6$

$$\therefore 6 = \frac{-10^2}{180} (1 + \tan^2 \theta) + 10 \tan \theta + 2$$

$$6 = \frac{-5}{9} (1 + \tan^2 \theta) + 10 \tan \theta + 2$$

$$54 = -5 - 5 \tan^2 \theta + 90 \tan \theta + 18$$

$$5 \tan^2 \theta - 90 \tan \theta + 41 = 0$$

$$\tan \theta = \frac{90 \pm \sqrt{90^2 - 4 \times 5 \times 41}}{10}$$

$$= \frac{90 \pm \sqrt{7280}}{10}$$

$$\theta = 87^\circ, 25^\circ \quad (\theta \text{ is acute})$$

2

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Suggested Solutions

Question 15

$$b) \quad y = 2 \sin^{-1}(2x-1) + \frac{\pi}{2}$$

$$\frac{y - \frac{\pi}{2}}{2} = \sin^{-1}(2x-1)$$

$$\text{Domain} : -1 \leq 2x-1 \leq 1$$

$$0 \leq x \leq 1$$

Range :

$$-\frac{\pi}{2} \leq \frac{y - \frac{\pi}{2}}{2} \leq \frac{\pi}{2}$$

$$-\pi \leq y - \frac{\pi}{2} \leq \pi$$

$$-\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$$

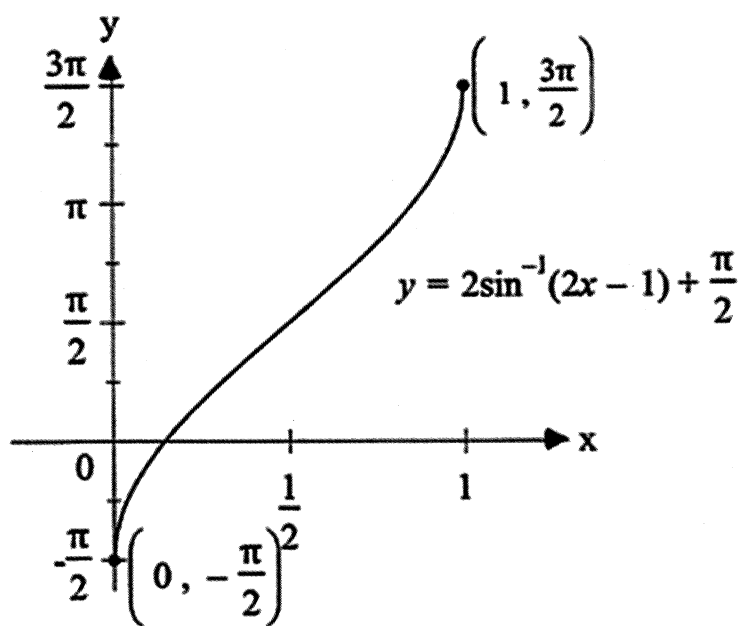
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Suggested Solutions

Question 15

b (ii)



2

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Suggested Solutions

Question 16

$$a) \sin 3x - \sin x + \cos 2x = 0; \quad 0 \leq x \leq 2\pi$$

$$\sin(2x+x) - \sin(2\pi-x) + \cos 2x = 0$$

$$2 \cos 2x \sin x + \cos 2x = 0 \quad \underline{\underline{!}}$$

$$\cos 2x (2 \sin x + 1) = 0$$

$$\cos 2x = 0 \quad ; \quad 0 \leq 2x \leq 4\pi$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \quad \underline{\underline{!}}$$

$$\text{or } \sin x = -\frac{1}{2} \quad \phi 3, 4$$

$$x = \frac{7\pi}{6}, \frac{11\pi}{6} \quad \underline{\underline{!}}$$

$$x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

3

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Suggested Solutions

Question 16

$$b) i) \quad 8 \cos^4 x = 3 + 4 \cos 2x + \cos 4x$$

$$\text{LHS} = 8 \cos^4 x$$

$$= 2 (2 \cos^2 x)^2$$

$$= 2 (1 + \cos 2x)^2$$

$$= 2 (1 + 2 \cos 2x + \cos^2 2x)$$

$$= 2 \left[1 + 2 \cos 2x + \frac{1}{2} (1 + \cos 4x) \right]$$

$$= 2 + 4 \cos 2x + 1 + \cos 4x$$

$$= 3 + 4 \cos 2x + \cos 4x$$

$$= \text{RHS}.$$

2

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Suggested Solutions

Question 16

b (ii)

$$V = \pi \int_0^4 \cos^4 x \, dx$$

$$= \pi \int_0^4 \frac{1}{8} (3 + 4 \cos 2x + \cos 4x) \, dx$$

$$= \frac{\pi}{8} \left[3x + 2 \sin 2x + \frac{1}{4} \sin 4x \right]_0^4 \quad \underline{\underline{1}}$$

$$= \frac{\pi}{8} \left[3\left(\frac{\pi}{4}\right) + 2 \sin\left(\frac{\pi}{2}\right) + \frac{1}{4} \sin \pi \right] - 0$$

$$V = \frac{\pi}{8} \left[\frac{3\pi}{4} + 2 \right] \pi^3 \quad \underline{\underline{1}}$$

2

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Suggested Solutions

Question 16

$$C. (i) {}^{13}C_5 = 1287$$

[1]

(ii) More women than men

$$\Rightarrow 5W : {}^7C_5 = 21$$

$$\Rightarrow 4W, 1M : {}^7C_4 \times {}^6C_1 = 210$$

$$\Rightarrow 3W, 2M : {}^7C_3 \times {}^6C_2 = 525$$

!

$$\text{Total} = 21 + 210 + 525$$

!

$$= 756$$

[2]

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